

2010.14265 — formula report

369 inline formulas, 16 display equations, 0 tables, 6 unrecovered image regions.

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2010.14265_E00012	012	■ 0.492 ● C +117	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(2): Z \perp_P W \mid \emptyset \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): Z \perp_P Y \mid \{W\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): Z \not\lhd_P X \mid \{W, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): W \not\lhd_P Y \mid \{Z\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): W \not\lhd_P X \mid \{Z, Y\} \\ & \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\lhd_P X \mid \{Z, W\} \end{aligned}$	(not rendered)	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\lhd_P Y \mid \{Z, W\} \\ & \pi(2) \rightarrow \pi(4): Z \not\lhd_P Y \mid \{X, W\} \\ & \pi(3) \rightarrow \pi(4): W \not\lhd_P Y \mid \{X, Z\} \end{aligned}$
2010.14265_E00009	011	■ 0.578 ● C +51	$\begin{aligned} & \pi(1) \rightarrow \pi(4): X \not\perp_P Y \mid \{Z, W\} \text{ \& } \pi(2) \rightarrow \pi(4): \\ & Z \not\perp_P Y \mid \{X, W\} \text{ \& } \pi(3) \rightarrow \pi(4): \\ & W \perp_P Y \mid \{X, Z\} \end{aligned}$	(not rendered)	$X_{\pi(j)} \not\lhd_P X_{\pi(k)} \mid \{X_{\pi(1)}, X_{\pi(2)}, \dots, X_{\pi(k-1)}\} \setminus \{X_{\pi(j)}\} \text{ ,}$
2010.14265_E00008	011	■ 0.675 ● C +34	$X_{\{\pi(j)\}} \not\perp_P X_{\{\pi(k)\}} \mid \left\{X_{\{\pi(1)\}}, X_{\{\pi(2)\}}, \ldots, X_{\{\pi(k-1)\}}\right\} \backslash \left\{X_{\{\pi(j)\}}\right\} \text{ ,}$	(not rendered)	$Z \perp_G \{X, Y\} \mid \text{Pa}(Z) \text{ .}$
2010.14265_E00013	013	■ 0.706 ● N +2	$Z \perp_G \{X, Y\} \mid \operatorname{Pa}(Z) \text{ .}$	(not rendered)	$\forall S \subseteq V \setminus \{X, Y, Z\} : A \not\lhd_P B \mid S \cup \{X, Y, Z\} \setminus \{A, B\} \text{ ,}$
2010.14265_E00001	004	■ 0.823 ● C +23	$\forall S \subseteq V \setminus \{X, Y, Z\} : A \not\perp_P B \mid S \cup \{X, Y, Z\} \setminus \{A, B\} \text{ ,}$	(not rendered)	$\begin{aligned} P(X = 1, W = 1, Y = 1, Z = 1) &= \frac{pq}{4} \text{ ,} \\ P(X = 1, Y = 1, Z = 1) &= \frac{q}{4} \text{ ,} \\ P(W = 1, Y = 1, Z = 1) &= \frac{p}{4} \text{ .} \end{aligned}$
2010.14265_E00006	011	■ 0.874 ● N +0	$\begin{aligned} & P(X=1, \text{ \& } W=1, Y=1, Z=1)=\frac{p \text{ } q}{4} \text{ , } \backslash \\ & \text{ \& } P(X=1, Y=1, Z=1)=\frac{q}{4} \text{ , } \backslash \text{ \& } P(W=1, Y=1, Z=1)= \\ & \frac{p}{4} \text{ . } \end{aligned}$	$\begin{aligned} & P(X = 1, W = 1, Y = 1, Z = 1) = \frac{pq}{4} \text{ ,} \\ & P(X = 1, Y = 1, Z = 1) = \frac{q}{4} \text{ ,} \\ & P(W = 1, Y = 1, Z = 1) = \frac{p}{4} \text{ .} \end{aligned}$	$X \perp_G Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}) \text{ ,}$
2010.14265_E00015	013	■ 0.960 ● C +14	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\}) \text{ ,}$	(not rendered)	$X \perp_G Z \mid \text{Pa}(X) \cup (X \setminus \{X\}) \cup (Z \setminus \{Z\}) \text{ ,}$
2010.14265_E00014 (1)	013	■ 0.980 ● C +14	$X \perp_G Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\}) \text{ ,}$	(not rendered)	

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
2010.14265_E00011	012	■ 0.998 ● C +99	$\begin{aligned} & \pi^{\prime}(1) \rightarrow \pi^{\prime}(3): X \not\perp_P Y \mid \{Z\} \quad \& \quad \pi^{\prime}(2) \rightarrow \pi^{\prime}(3): Z \not\perp_P Y \mid \{X\} \\ & Y \mid \{X\} \quad \& \quad \pi^{\prime}(3) \rightarrow \pi^{\prime}(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi^{\prime}(1) \rightarrow \pi^{\prime}(4): X \not\perp_P W \mid \{Z, Y\} \\ & \pi^{\prime}(2) \rightarrow \pi^{\prime}(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$	(not rendered)	$\begin{aligned} & \pi'(1) \rightarrow \pi'(3): X \not\perp_P Y \mid \{Z\} \\ & \pi'(2) \rightarrow \pi'(3): Z \not\perp_P Y \mid \{X\} \\ & \pi'(3) \rightarrow \pi'(4): Y \not\perp_P W \mid \{X, Z\} \\ & \pi'(1) \rightarrow \pi'(4): X \not\perp_P W \mid \{Z, Y\} \\ & \pi'(2) \rightarrow \pi'(4): Z \not\perp_P W \mid \{X, Y\} \end{aligned}$
2010.14265_E00010	012	■ 1.000 ● C +39	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X . \end{aligned}$	(not rendered)	$\begin{aligned} & \pi(1) \rightarrow \pi(2): X \perp_P Z \mid \emptyset \\ & \pi(1) \rightarrow \pi(3): X \perp_P W \mid Z \\ & \pi(2) \rightarrow \pi(3): Z \perp_P W \mid X . \end{aligned}$
2010.14265_E00016	013	■ 1.000 ● C +14	$X \perp_P Z \mid \boldsymbol{S} \cup \{Y\} \cup (X \backslash \{X\}) \cup (Z \backslash \{Z\}) .$	(not rendered)	$X \perp_P Z \mid \boldsymbol{S} \cup \{Y\} \cup (X \backslash \{X\}) \cup (Z \backslash \{Z\}) .$
2010.14265_E00002	006	■ 1.000 ● K +0	$Y := ((X \oplus Z) \wedge W) \oplus E,$	$Y := ((X \oplus Z) \wedge W) \oplus E,$	$Y := ((X \oplus Z) \wedge W) \oplus E ,$
2010.14265_E00003	011	■ 1.000 ● K +0	$Y := ((X \oplus Z) \wedge W) \oplus E .$	$Y := ((X \oplus Z) \wedge W) \oplus E.$	$Y := ((X \oplus Z) \wedge W) \oplus E .$
2010.14265_E00004	011	■ 1.000 ● W +0	$\begin{aligned} & P(X=0, Z=1, Y=1) \quad \& \quad = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p + q - 2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p - pq = \frac{p}{2} . \end{aligned}$	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p + q - 2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p - pq = \frac{p}{2} . \end{aligned}$	$\begin{aligned} & P(X=0, Z=1, Y=1) = P(X=0, Z=1) \cdot P(Y=1) \\ & \Leftrightarrow p + q - 2pq = q + \frac{p}{2} - pq \\ & \Leftrightarrow p - pq = \frac{p}{2} . \end{aligned}$
2010.14265_E00005	011	■ 1.000 ● N +0	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)} .$	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)} .$	$P(X, W \mid Y, Z) = \frac{P(X, W, Y, Z)}{P(Y, Z)} .$
2010.14265_E00007	011	■ 1.000 ● N +0	$p \, q = \frac{p \, q}{2 P(Y=1)} .$	$pq = \frac{pq}{2 P(Y=1)} .$	$pq = \frac{pq}{2 P(Y=1)} .$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					

Inline formulas (first occurrence)

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00001	001	—	$\{ \ }^{\wedge\{1,2\}}$	$1,2$
2010.14265_F00002	001	—	P	P
2010.14265_F00003	001	—	G	G
2010.14265_F00004	001	—	$X, \ Z$	X, Z
2010.14265_F00005	001	—	$Y:=(X \ \operatorname{oplus} \ Z) \ \operatorname{oplus} \ E$	$Y := (X \oplus Z) \oplus E$
2010.14265_F00006	001	—	X	X
2010.14265_F00007	001	—	Y	Y
2010.14265_F00008	001	—	$X \ \operatorname{Perp}_{\{P\}} \ Y$	<i>(not rendered)</i>
2010.14265_F00009	001	—	$Z \ \operatorname{Perp}_{\{P\}} \ Y$	<i>(not rendered)</i>
2010.14265_F00010	001	—	$X \ \operatorname{Perp}_{\{P\}} \ Z$	<i>(not rendered)</i>
2010.14265_F00011	001	—	$Z \ \operatorname{not} \ \operatorname{Perp}_{\{P\}} \ Y$	<i>(not rendered)</i>
2010.14265_F00012	001	—	Z	Z
2010.14265_F00013	001	—	$X, \ Y$	X, Y
2010.14265_F00014		—	$37\{ \ }^{\wedge\{\operatorname{text}\{th\}\}}$	37^{th}
2010.14265_F00015	002	—	$\operatorname{\boldsymbol{V}}$	$\boldsymbol{V}$
2010.14265_F00016	002	—	$X \ \rightarrowtail Y$	$X \rightarrow Y$
2010.14265_F00017	002	—	$X, \ Y \ \operatorname{in} \ \operatorname{\boldsymbol{V}}$	$X, Y \in \boldsymbol{V}$
2010.14265_F00018	002	—	$X \ \operatorname{in} \ \operatorname{\boldsymbol{V}}$	$X \in \boldsymbol{V}$
2010.14265_F00019	002	—	$\operatorname{Pa}(X)$	$\operatorname{Pa}(X)$
2010.14265_F00020	002	—	$\operatorname{Ch}(X)$	$\operatorname{Ch}(X)$
2010.14265_F00021	002	—	$\operatorname{PC}(X) := \operatorname{Pa}(X) \ \operatorname{cup} \ \operatorname{Ch}(X)$	$\operatorname{PC}(X) := \operatorname{Pa}(X) \cup \operatorname{Ch}(X)$
2010.14265_F00022	002	—	$\operatorname{An}(X)$	$\operatorname{An}(X)$
2010.14265_F00023	002	—	$\operatorname{De}(X)$	$\operatorname{De}(X)$
2010.14265_F00024	002	—	$\operatorname{Nd}(X) := \operatorname{\boldsymbol{V}} \ \backslash \ \operatorname{De}(X)$	$\operatorname{Nd}(X) := \boldsymbol{V} \setminus \operatorname{De}(X)$
2010.14265_F00025	002	—	$\operatorname{MB}(X) :=$	$\operatorname{MB}(X) :=$
2010.14265_F00026	002	—	$\operatorname{PC}(X) \ \operatorname{cup} \ \operatorname{Sp}(X)$	$\operatorname{PC}(X) \cup \operatorname{Sp}(X)$
2010.14265_F00027	002	—	$\operatorname{Sp}(X)$	$\operatorname{Sp}(X)$
2010.14265_F00028	002	—	d	d
2010.14265_F00029	002	—	$\operatorname{MB}(X)$	$\operatorname{MB}(X)$
2010.14265_F00030	002	—	$Z \ \operatorname{notin} \ \operatorname{\boldsymbol{V}}$	$Z \notin \boldsymbol{V}$
2010.14265_F00031	002	—	$\operatorname{\mathcal{M}}=(G, \ \operatorname{\boldsymbol{V}}, \ P)$	$\mathcal{M} = (G, \boldsymbol{V}, P)$
2010.14265_F00032	002	—	$\operatorname{\boldsymbol{X}}, \ \operatorname{\boldsymbol{Y}}, \ \operatorname{\boldsymbol{Z}} \ \operatorname{subseq} \ \operatorname{\boldsymbol{V}}$	$\boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$
2010.14265_F00033	002	—	$\operatorname{\boldsymbol{X}}$	$\boldsymbol{X}$
2010.14265_F00034	002	—	$\operatorname{\boldsymbol{Y}}$	$\boldsymbol{Y}$
2010.14265_F00035	002	—	$\operatorname{\boldsymbol{Z}}$	$\boldsymbol{Z}$
2010.14265_F00036	002	—	$\operatorname{\boldsymbol{X}} \ \operatorname{Perp}_{\{P\}} \ \operatorname{\boldsymbol{Y}} \ \operatorname{mid} \ \operatorname{\boldsymbol{Z}}$	<i>(not rendered)</i>
2010.14265_F00037	002	—	D	D
2010.14265_F00038	002	—	p	p
2010.14265_F00039	002	—	$p=\angle X, \ \ldots, \ Y\angle$	$p = \langle X, \dots, Y \rangle$
2010.14265_F00040	002	—	$X_{\{1\}}, \ \ldots, \ X_{\{n\}}$	$X_1, \dots, X_n$
2010.14265_F00041	002	—	$X_{\{i\}}$	X_i
2010.14265_F00042	002	—	$X_{\{i+1\}}$	X_{i+1}
2010.14265_F00043	002	—	$i=1, \ \ldots, \ n-1, \ X_{\{1\}}=X$	$i = 1, \dots, n-1, X_1 = X$
2010.14265_F00044	002	—	$X_{\{n\}}=Y$	$X_n = Y$
2010.14265_F00045	002	—	C	C
2010.14265_F00046	002	—	$\angle\ldots, \ X, \ C, \ Y, \ \ldots\angle$	$\langle \dots, X, C, Y, \dots \rangle$
2010.14265_F00047	002	—	$X \ \rightarrowtail C \ \leftarrowtail Y$	$X \rightarrow C \leftarrow Y$
2010.14265_F00048		—	$\operatorname{\mathbf{Z}}$	$\boldsymbol{Z}$
2010.14265_F00049	002	—	$X \ \operatorname{in} \ \operatorname{\boldsymbol{X}}$	$X \in \boldsymbol{X}$
2010.14265_F00050	002	—	$Y \ \operatorname{in} \ \operatorname{\boldsymbol{Y}}, \ X$	$Y \in \boldsymbol{Y}, X$
2010.14265_F00051	002	—	$\operatorname{\boldsymbol{X}} \ \operatorname{Perp}_{\{G\}} \ \operatorname{\boldsymbol{Y}} \ \operatorname{mid} \ \operatorname{\boldsymbol{Z}}$	<i>(not rendered)</i>
2010.14265_F00052	003	—	$\operatorname{\mathcal{M}}=$	$\mathcal{M} =$
2010.14265_F00053	003	—	$(G, \ \operatorname{\boldsymbol{V}}, \ P)$	$(G, \boldsymbol{V}, P)$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value				
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00054	003	—	$\backslash\boldsymbol{S} \subseteq \backslash\boldsymbol{V} \backslash\{X, Y\}$	$\boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y\}$
2010.14265_F00055	003	—	$G, \backslash\boldsymbol{V}, P$	$G, \boldsymbol{V}, P$
2010.14265_F00056	003	—	$\langle X, Y, Z \rangle$	$\langle X, Y, Z \rangle$
2010.14265_F00057	003	—	$X \rightarrowtail Y \leftarrowtail Z$	$X \rightarrow Y \leftarrow Z$
2010.14265_F00058	003	—	$V \backslash\{X, Z\}$	$V \backslash \{X, Z\}$
2010.14265_F00059	003	—	$X, Y, Z \in \backslash\boldsymbol{V}$	$X, Y, Z \in \boldsymbol{V}$
2010.14265_F00060		—	$\langle X, Y, Z \rangle, X$	$\langle X, Y, Z \rangle, X$
2010.14265_F00061	003	—	$X, Z \sim N(0,1)$	$X, Z \sim N(0, 1)$
2010.14265_F00062	003	—	$Y := \operatorname{sign}(X\,Z) \cdot E$	$Y := \operatorname{sign}(XZ) \cdot E$
2010.14265_F00063	003	—	$E \sim$	$E \sim$
2010.14265_F00064	003	—	$\operatorname{Exp}\!\left(\frac{1}{\sqrt{2}}\right)$	$\operatorname{Exp}\!\left(\frac{1}{\sqrt{2}}\right)$
2010.14265_F00065	003	—	$\mathcal{M} = (G, V, P)$	$\mathcal{M} = (G, V, P)$
2010.14265_F00066	003	—	$X \oplus Z :=$	$X \oplus Z :=$
2010.14265_F00067	003	—	$(X+Z) \bmod 2$	$(X + Z) \bmod 2$
2010.14265_F00068	003	—	E	E
2010.14265_F00069	003	—	$P(E=1)=p$	$P(E = 1) = p$
2010.14265_F00070	003	—	$\theta \leq p < \frac{1}{2}$	$0 \leq p < \frac{1}{2}$
2010.14265_F00071	003	—	$E \perp_P \backslash\{X, Z\}$	<i>(not rendered)</i>
2010.14265_F00072	003	—	$X \perp_G \{Y, Z\} \perp_G Y$	<i>(not rendered)</i>
2010.14265_F00073	003	—	$Y \not\perp_P \backslash\{X, Z\}$	<i>(not rendered)</i>
2010.14265_F00074	003	—	$X \perp_P \{Z\} \mid Y$	<i>(not rendered)</i>
2010.14265_F00075	003	—	$P(X=1, Z=1, Y=1)=\frac{p}{4}$	$P(X = 1, Z = 1, Y = 1) = \frac{p}{4}$
2010.14265_F00076	003	—	$P(X=1, Z=1) \cdot P(Y=1)=\frac{1}{4} \cdot \frac{1}{2}=\frac{1}{8}$	$P(X = 1, Z = 1) \cdot P(Y = 1) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$
2010.14265_F00077	003	—	$p=\frac{1}{2}$	$p = \frac{1}{2}$
2010.14265_F00078	003	—	$E \sim \operatorname{Exp}\!\left(\frac{1}{\sqrt{2}}\right)$	$E \sim \operatorname{Exp}\!\left(\frac{1}{\sqrt{2}}\right)$
2010.14265_F00079	004	—	(X)	(X)
2010.14265_F00080	004	—	(Y)	(Y)
2010.14265_F00081	004	—	$Z := \alpha X, Y := \beta Z - \gamma X$	$Z := \alpha X, Y := \beta Z - \gamma X$
2010.14265_F00082	004	—	$\gamma = \alpha \beta$	$\gamma = \alpha \beta$
2010.14265_F00083	004	—	$X \rightarrowtail Z \leftarrowtail Y$	$X \rightarrow Z \leftarrow Y$
2010.14265_F00084	004	—	$Z \rightarrowtail W \rightarrowtail Y$	$Z \rightarrow W \rightarrow Y$
2010.14265_F00085	004	—	$G^{\prime}$	G'
2010.14265_F00086	004	—	$\{X, Y, Z\}$	$\{X, Y, Z\}$
2010.14265_F00087	004	—	P^3	P^3
2010.14265_F00088	004	—	$A, B \in \{X, Y, Z\}$	$A, B \in \{X, Y, Z\}$
2010.14265_F00089		—	$X \nvdash_P \{Y, Z\}$	$X \not\vdash_P \{Y, Z\}$
2010.14265_F00090		—	$Y \nvdash_P \{X, Z\}$	$Y \not\vdash_P \{X, Z\}$
2010.14265_F00091		—	$Z \not\measuredangle_P \{X, Y\}$	$Z \not\llcorner_P \{X, Y\}$
2010.14265_F00092	005	—	U	U
2010.14265_F00093	005	—	$\{U, Y, Z\}$	$\{U, Y, Z\}$
2010.14265_F00094	005	—	$\{W, Z\}$	$\{W, Z\}$
2010.14265_F00095	005	—	$\{U, W, Z\}$	$\{U, W, Z\}$
2010.14265_F00096	005	—	$X \nVdash_P \{Y, Z\}$	$X \not\Vdash_P \{Y, Z\}$
2010.14265_F00097	005	—	$Y \nVdash_P \{X, Z\}$	$Y \not\Vdash_P \{X, Z\}$
2010.14265_F00098	005	—	$Z \not\perp_P \backslash\{X, Y\}$	<i>(not rendered)</i>
2010.14265_F00099	005	—	$\{X, Y, Z\} \subseteq \backslash\boldsymbol{V}$	$\{X, Y, Z\} \subseteq \boldsymbol{V}$
2010.14265_F00100	005	—	$X \perp_P \{Y, Z\}, Y \not\perp_P \backslash\{X, Z\}$	<i>(not rendered)</i>
2010.14265_F00101	005	—	$Z \perp_P \{X, Y\}$	<i>(not rendered)</i>
2010.14265_F00102	005	—	$X \perp_P \{Y\} \mid Z$	<i>(not rendered)</i>
2010.14265_F00103	005	—	$Y, \measuredangle_P Z \mid X$	$Y, \angle_P Z \mid X$
2010.14265_F00104	005	—	X, Y, Z	X, Y, Z
2010.14265_F00105	005	—	$C\,M\,C$	CMC
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00106	005	—	$X, Y, Z \in V$	$X, Y, Z \in V$
2010.14265_F00107	005	—	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y, Z\}$	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y, Z\}$
2010.14265_F00108	005	—	$X, \not\perp_P Y \left Z \cup S, X, \perp_P Z \right Y \cup S$	<i>(not rendered)</i>
2010.14265_F00109	005	—	$Y, \perp_P Z \mid X \cup S$	<i>(not rendered)</i>
2010.14265_F00110	005	—	k	k
2010.14265_F00111	005	—	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y\} : X \nVdash_P Y \mid \boldsymbol{S}$	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y\} : X \not\vdash_P Y \mid \boldsymbol{S}$
2010.14265_F00112	005	—	$X, Y_{\{1\}}, Y_{\{2\}} \in \boldsymbol{V}, X$	$X, Y_1, Y_2 \in \boldsymbol{V}, X$
2010.14265_F00113	005	—	$\left\{ Y_{\{1\}}, Y_{\{2\}} \right\}$	$\{Y_1, Y_2\}$
2010.14265_F00114	005	—	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y_{\{1\}}, Y_{\{2\}}\}$	$\forall \boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Y_1, Y_2\}$
2010.14265_F00115		—	$X \nVdash_P Y_{\{1\}} \mid \boldsymbol{S} \cup Y_{\{2\}}$	$X \not\vdash_P Y_1 \mid \boldsymbol{S} \cup Y_2$
2010.14265_F00116		—	$X \nvdash_P Y_{\{2\}} \mid \boldsymbol{S} \cup Y_{\{1\}}$	$X \not\vdash_P Y_2 \mid \boldsymbol{S} \cup Y_1$
2010.14265_F00117		—	$Y_{\{1\}} \not\perp \}_P Y_{\{2\}} \mid \boldsymbol{S} \cup X$	<i>(not rendered)</i>
2010.14265_F00118	005	—	$Y_{\{1\}}$	Y_1
2010.14265_F00119	005	—	$Y_{\{2\}}$	Y_2
2010.14265_F00120	005	—	$X_{\{2\}} \backslash \{Y, Z\}$	$X_{-2} \{Y, Z\}$
2010.14265_F00121	005	—	$X_{\{ \}_1} Y$	$X_{-1} Y$
2010.14265_F00122	005	—	$\{ \}_{\{2\}}$	2
2010.14265_F00123	005	—	$X_{\leq 2} \boldsymbol{Y}$	$X_{-\leq 2} \boldsymbol{Y}$
2010.14265_F00124	005	—	$X \stackrel{s}{\sim}_{\leq 2} \boldsymbol{Y}$	$X \stackrel{s}{-}_{\leq 2} \boldsymbol{Y}$
2010.14265_F00125	005	—	$X \rightarrow Y \rightarrow Z$	$X \rightarrow Y \rightarrow Z$
2010.14265_F00126	005	—	$X_{\{1\}} Y$	$X_{-1} Y$
2010.14265_F00127	005	—	$Y_{\{ \}_1} Z$	$Y_{-1} Z$
2010.14265_F00128	005	—	$X \stackrel{s}{\sim}_{\{2\}} \{Y, Z\}, Y \stackrel{s}{\sim}_{\{2\}} \{X, Z\}$	$X \stackrel{s}{-}_{-2} \{Y, Z\}, Y \stackrel{s}{-}_{-2} \{X, Z\}$
2010.14265_F00129	006	—	$Z \frac{\{s\}_{\{2\}}}{\{X, Y\}}$	$Z \frac{s}{22} \{X, Y\}$
2010.14265_F00130	006	—	$\boldsymbol{Y} \subseteq M B(X)$	$\boldsymbol{Y} \subseteq MB(X)$
2010.14265_F00131	006	—	$Y \in \boldsymbol{Y}$	$Y \in \boldsymbol{Y}$
2010.14265_F00132	006	—	$X^{\{s\}_{\{ \}_{\leq 2}}} \boldsymbol{Y}$	$X \stackrel{s}{<}_2 \boldsymbol{Y}$
2010.14265_F00133	006	—	$Y \stackrel{s}{\sim}_{\{2\}} \{X, Z\}$	$Y \stackrel{s}{-}_{-2} \{X, Z\}$
2010.14265_F00134	006	—	$X \frac{\{s\}_{\{2\}}}{\{Y, Z\}}$	$X \frac{s}{22} \{Y, Z\}$
2010.14265_F00135	006	—	$X \rightarrow Z \rightarrow W \rightarrow Y$	$X \rightarrow Z \rightarrow W \rightarrow Y$
2010.14265_F00136	006	—	$X \perp_P Y, X \perp_P W \mid Z, X$	<i>(not rendered)</i>
2010.14265_F00137	006	—	W	W
2010.14265_F00138	006	—	$\{W, Y\}$	$\{W, Y\}$
2010.14265_F00139	006	—	$Y \frac{\{s\}_{\{-\}_{\{2\}}}}{\{X, Z\}}$	$Y \frac{s}{-2} \{X, Z\}$
2010.14265_F00140	006	—	$Y_{\{ \}_1} W$	$Y_{-1} W$
2010.14265_F00141	006	—	$\{X, Z\}$	$\{X, Z\}$
2010.14265_F00142	006	—	$P(W=1)=p$	$P(W=1)=p$
2010.14265_F00143	006	—	$0<p<1$	$0 < p < 1$
2010.14265_F00144	006	—	$P(E=1)=q$	$P(E=1)=q$
2010.14265_F00145	006	—	$0<q<\frac{1}{2}$	$0 < q < \frac{1}{2}$
2010.14265_F00146	006	—	E, W, X	E, W, X
2010.14265_F00147	006	—	$(q \neq 0)$	$(q \neq 0)$
2010.14265_F00148	006	—	$\left(q \neq \frac{1}{2}\right)$	$\left(q \neq \frac{1}{2}\right)$
2010.14265_F00149	006	—	$\{X, Z\} \perp_P W$	<i>(not rendered)</i>
2010.14265_F00150	006	—	$X \perp_P W \mid Y$	<i>(not rendered)</i>
2010.14265_F00151	006	—	$Z \perp_P W \mid Y$	<i>(not rendered)</i>
2010.14265_F00152	006	—	$X \perp_P W \mid \{Y, Z\}$	<i>(not rendered)</i>
2010.14265_F00153	006	—	$Z, \perp_P W \mid \{Y, X\}$	<i>(not rendered)</i>
2010.14265_F00154	006	—	$W \rightarrow Y$	$W \rightarrow Y$
2010.14265_F00155	006	—	$X \nVdash_P W \mid Z$	$X \not\vdash_P W \mid Z$
2010.14265_F00156	006	—	$\boldsymbol{Y} \rightarrow X$	$\boldsymbol{Y} \rightarrow X$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00157	006	—	$Y \rightarrow X$	$Y \rightarrow X$
2010.14265_F00158	006	—	$X \rightarrow \boldsymbol{Y}$	$X \rightarrow \boldsymbol{Y}$
2010.14265_F00159	006	—	$\forall Y \in \boldsymbol{Y}: X \rightarrow Y$	$\forall Y \in \boldsymbol{Y}: X \rightarrow Y$
2010.14265_F00160	006	—	$M := (G, V, P)$	$M := (G, V, P)$
2010.14265_F00161	006	—	$\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	$\boldsymbol{X}, \boldsymbol{Z} \subseteq \boldsymbol{V}$
2010.14265_F00162	006	—	$Y \in \boldsymbol{V}$	$Y \in \boldsymbol{V}$
2010.14265_F00163	007	—	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{X}$	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{X}$
2010.14265_F00164	007	—	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{Z}$	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{Z}$
2010.14265_F00165	007	—	$Z \in Z$	$Z \in Z$
2010.14265_F00166	007	—	$Z \in \boldsymbol{Z}$	$Z \in \boldsymbol{Z}$
2010.14265_F00167	007	—	$Y \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	$Y \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$
2010.14265_F00168	007	—	$\boldsymbol{X} \rightarrow Y \leftarrow \boldsymbol{Z}$	$\boldsymbol{X} \rightarrow Y \leftarrow \boldsymbol{Z}$
2010.14265_F00169	007	—	$\boldsymbol{V} \setminus \{X, Z\}$	$\boldsymbol{V} \setminus \{X, Z\}$
2010.14265_F00170	007	—	$(\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	$(\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$
2010.14265_F00171	007	—	Y, Y	Y, Y
2010.14265_F00172	007	—	$X \leftarrow Y \rightarrow Z$	$X \leftarrow Y \rightarrow Z$
2010.14265_F00173	007	—	$Y \frac{s}{2_2} \{X, Z\}, Y -_{-1} W$	$Y \frac{s}{2_2} \{X, Z\}, Y -_{-1} W$
2010.14265_F00174	007	—	$\{Y, Z\}$	$\{Y, Z\}$
2010.14265_F00175	007	—	$\{Y, X\}$	$\{Y, X\}$
2010.14265_F00176	007	—	$\{X, Z\} \rightarrow Y \leftarrow W$	$\{X, Z\} \rightarrow Y \leftarrow W$
2010.14265_F00177	007	—	$X \not\perp_P Z$	<i>(not rendered)</i>
2010.14265_F00178	007	—	$X \nVdash_P Y$	$X \nVdash_P Y$
2010.14265_F00179	007	—	$Y, \measuredangle_P Z$	$Y, \angle_P Z$
2010.14265_F00180	007	—	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{X}$	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{X}$
2010.14265_F00181	007	—	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{Z}$	$Y \stackrel{s}{\rightarrow}_{\leq 2} \boldsymbol{Z}$
2010.14265_F00182	007	—	$\boldsymbol{Z} \in \boldsymbol{Z}$	$\boldsymbol{Z} \in \boldsymbol{Z}$
2010.14265_F00183	007	—	$Y \stackrel{s}{\rightarrow}_{-2} \boldsymbol{X} = \{X_1, X_2\}, Y \stackrel{s}{\rightarrow}_{-2} \boldsymbol{Z} =$	$Y \stackrel{s}{\rightarrow}_{-2} \boldsymbol{X} = \{X_1, X_2\}, Y \stackrel{s}{\rightarrow}_{-2} \boldsymbol{Z} =$
2010.14265_F00184	007	—	$\{Z_1, Z_2\}$	$\{Z_1, Z_2\}$
2010.14265_F00185	007	—	$X_2^s \{U, Z_2\}$	$X_2^s \{U, Z_2\}$
2010.14265_F00186	007	—	$\langle X_2, Y, Z_2 \rangle$	$\langle X_2, Y, Z_2 \rangle$
2010.14265_F00187	007	—	X_1	X_1
2010.14265_F00188	007	—	Z_2	Z_2
2010.14265_F00189	007	—	X_2	X_2
2010.14265_F00190	007	—	$X, Z \in \boldsymbol{X} \times \boldsymbol{Z}$	$X, Z \in \boldsymbol{X} \times \boldsymbol{Z}$
2010.14265_F00191	007	—	$\{Y\} \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$	$\{Y\} \cup (\boldsymbol{X} \setminus \{X\}) \cup (\boldsymbol{Z} \setminus \{Z\})$
2010.14265_F00192	007	—	$Z \in Z, X$	$Z \in Z, X$
2010.14265_F00193	007	—	$Z \in \boldsymbol{Z}, X$	$Z \in \boldsymbol{Z}, X$
2010.14265_F00194	007	—	$(X, Z) \in \boldsymbol{X} \times \boldsymbol{Z}$	$(X, Z) \in \boldsymbol{X} \times \boldsymbol{Z}$
2010.14265_F00195	008	—	$\{Z, U\}$	$\{Z, U\}$
2010.14265_F00196	008	—	$X \rightarrow U \leftarrow Z$	$X \rightarrow U \leftarrow Z$
2010.14265_F00197	008	—	$M := (G, \boldsymbol{V}, P)$	$M := (G, \boldsymbol{V}, P)$
2010.14265_F00198	008	—	$\boldsymbol{X}, \boldsymbol{Z} \subseteq$	$\boldsymbol{X}, \boldsymbol{Z} \subseteq$
2010.14265_F00199	008	—	$\boldsymbol{X} \cap \boldsymbol{Z} = \emptyset, Y_{-\leq 2} \boldsymbol{X}, Y_{-\leq 2}^s \boldsymbol{Z}$	$\boldsymbol{X} \cap \boldsymbol{Z} = \emptyset, Y_{-\leq 2} \boldsymbol{X}, Y_{-\leq 2}^s \boldsymbol{Z}$
2010.14265_F00200	008	—	$(X, Z) \in X \times Z$	$(X, Z) \in X \times Z$
2010.14265_F00201	008	—	$\{\boldsymbol{X}, Y, \boldsymbol{Z}\}$	$\{\boldsymbol{X}, Y, \boldsymbol{Z}\}$
2010.14265_F00202	008	—	T	T
2010.14265_F00203	008	—	$T \rightarrow C \leftarrow S$	$T \rightarrow C \leftarrow S$
2010.14265_F00204	008	—	$T -_{-1} Y$	$T -_{-1} Y$
2010.14265_F00205	008	—	$C -_{-1} Y$	$C -_{-1} Y$
2010.14265_F00206	008	—	$T \not\perp S \mid C$	<i>(not rendered)</i>
2010.14265_F00207	008	—	$\boldsymbol{X} = \{X, T\}$	$\boldsymbol{X} = \{X, T\}$
2010.14265_F00208	008	—	$\mathrm{PC}(T)$	$\mathrm{PC}(T)$
2010.14265_F00209	008	—	$\langle T, Y, Z \rangle$	$\langle T, Y, Z \rangle$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00210	008	—	$T \rightarrowtail Y \leftarrowtail Z$	$T \rightarrow Y \leftarrow Z$
2010.14265_F00211	008	—	$Y \leftarrowtail X \rightarrowtail Z$	$Y \leftarrow X \rightarrow Z$
2010.14265_F00212	008	—	$T_{-_{\{2\}}^{\{s\}}}\{X, Z\}$	$T -_2^s \{X, Z\}$
2010.14265_F00213	008	—	$T_{.\{P\}} X \mid \boldsymbol{S} \backslash X$	$T_{.P} X \mid \boldsymbol{S} \backslash X$
2010.14265_F00214	008	—	$Z \in \boldsymbol{S}$	$Z \in \boldsymbol{S}$
2010.14265_F00215	008	—	$\boldsymbol{S}$	$\boldsymbol{S}$
2010.14265_F00216	008	—	$M=(G, V, P)$	$M=(G, V, P)$
2010.14265_F00217	008	—	$M\,B(T)$	$MB(T)$
2010.14265_F00218	008	—	$T \in \boldsymbol{V}$	$T \in \boldsymbol{V}$
2010.14265_F00219	014	—	$\mathrm{MB}(T)$	$\mathbf{MB}(T)$
2010.14265_F00220		—	$\boldsymbol{V}^{\prime} \leftarrow \boldsymbol{V} \backslash \{T\}$	$\boldsymbol{V}' \leftarrow \boldsymbol{V} \backslash \{T\}$
2010.14265_F00221		—	$\boldsymbol{S} \leftarrow \emptyset$	$\boldsymbol{S} \leftarrow \emptyset$
2010.14265_F00222		—	$\left(\exists X \in \boldsymbol{V}^{\prime} : T \nVdash_P X \mid \boldsymbol{S}\right) \vee$	$(\exists X \in \boldsymbol{V}' : T \nVdash_P X \mid \boldsymbol{S}) \vee$
2010.14265_F00223		—	$\left(\exists X, Z \in \boldsymbol{V}^{\prime} : T \nVdash_P X \mid \boldsymbol{S} \cup \{Z\}\right)$	$(\exists X, Z \in \boldsymbol{V}' : T \nVdash_P X \mid \boldsymbol{S} \cup \{Z\})$
2010.14265_F00224		—	$\boldsymbol{S} \leftarrow \boldsymbol{S} \cup \{X\} ;$	$\boldsymbol{S} \leftarrow \boldsymbol{S} \cup \{X\};$
2010.14265_F00225		—	$\exists X \in \boldsymbol{S} : T \perp_P X \mid \boldsymbol{S} \backslash X$	<i>(not rendered)</i>
2010.14265_F00226		—	$\boldsymbol{S} \leftarrow \boldsymbol{S} \backslash X ;$	$\boldsymbol{S} \leftarrow \boldsymbol{S} \backslash X;$
2010.14265_F00227	009	—	$X \rightarrowtail Y \leftarrowtail Z, Y \rightarrowtail X \leftarrowtail Z$	$X \rightarrow Y \leftarrow Z, Y \rightarrow X \leftarrow Z$
2010.14265_F00228	009	—	$X \rightarrowtail Z$	$X \rightarrow Z$
2010.14265_F00229	011	—	$W \perp_P Y$	<i>(not rendered)</i>
2010.14265_F00230	011	—	X, Z, W	X, Z, W
2010.14265_F00231	011	—	$q>0$	$q>0$
2010.14265_F00232	011	—	$P(Y=1)=q+\frac{p}{2}-pq$	$P(Y=1)=q+\frac{p}{2}-pq$
2010.14265_F00233	011	—	$P(X=1, Y=1)=\frac{1}{2} \; P(Y=1)=P(X=$	$P(X=1, Y=1)=\frac{1}{2} P(Y=1)=P(X=$
2010.14265_F00234	011	—	$1) \cdot P(Y=1)$	$1) \cdot P(Y=1)$
2010.14265_F00235	011	—	$P(X=1, Y=0)=P(X=1) \cdot P(Y=0)$	$P(X=1, Y=0)=P(X=1) \cdot P(Y=0)$
2010.14265_F00236	011	—	$P(X=0, Z=1, Y=1)=$	$P(X=0, Z=1, Y=1)=$
2010.14265_F00237	011	—	$\frac{p+q-2pq}{4}$	$\frac{p+q-2pq}{4}$
2010.14265_F00238	011	—	$P(X=0, Z=1) \cdot P(Y=1)=\frac{1}{4} \; P(Y=1)$	$P(X=0, Z=1) \cdot P(Y=1)=\frac{1}{4} P(Y=1)$
2010.14265_F00239	011	—	$p=0$	$p=0$
2010.14265_F00240	011	—	$q=\frac{1}{2}$	$q=\frac{1}{2}$
2010.14265_F00241	011	—	$Y \not\perp_P X \mid Z$	<i>(not rendered)</i>
2010.14265_F00242	011	—	$Y \not\perp_P Z \mid X$	<i>(not rendered)</i>
2010.14265_F00243	011	—	$W \mathscr{H}_P Y$	$W \mathscr{H}_P Y$
2010.14265_F00244	011	—	$P(W=1, Y=1)=\frac{p}{2}$	$P(W=1, Y=1)=\frac{p}{2}$
2010.14265_F00245	011	—	$P(W=1) \cdot P(Y=1)$	$P(W=1) \cdot P(Y=1)$
2010.14265_F00246	011	—	$p=0, \; p=1$	$p=0, p=1$
2010.14265_F00247	011	—	$X \measuredangle_P W \mid \{Y, Z\}$	$X \angle_P W \mid \{Y, Z\}$
2010.14265_F00248	011	—	$Z \not\perp_P W \mid \{X, Y\}$	<i>(not rendered)</i>
2010.14265_F00249	011	—	$P(X \mid Y, Z) \cdot P(W \mid Y, Z)$	$P(X \mid Y, Z) \cdot P(W \mid Y, Z)$
2010.14265_F00250	011	—	$P(Y=1, Z=1)=P(Y=1) \; / \; 2$	$P(Y=1, Z=1)=P(Y=1)/2$
2010.14265_F00251	011	—	$q=0$	$q=0$
2010.14265_F00252	011	—	$p=1$	$p=1$
2010.14265_F00253	011	—	$X \perp_P W$	<i>(not rendered)</i>
2010.14265_F00254	011	—	$G^{\ast}$	$G^{\ast}$
2010.14265_F00255	011	—	G_{π}	G_{π}
2010.14265_F00256	011	—	π	π
2010.14265_F00257	011	—	E_{π}	E_{π}
2010.14265_F00258	011	—	j	j
2010.14265_F00259	011	—	$\pi(j)$	$\pi(j)$
2010.14265_F00260	011	—	$\pi(k)$	$\pi(k)$
2010.14265_F00261	011	—	$j<k$	$j<k$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5\text{--}0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

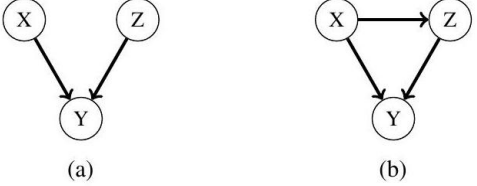
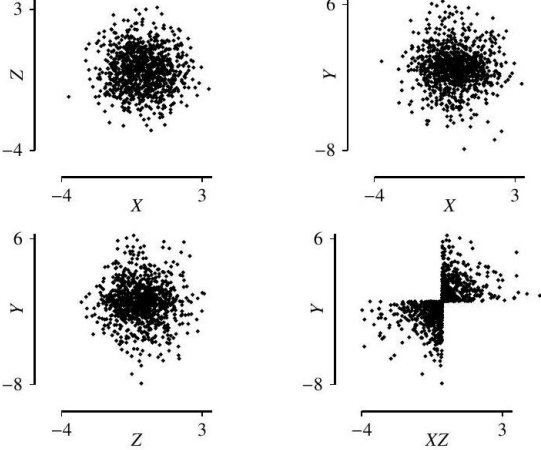
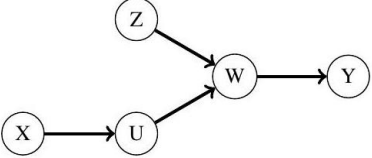
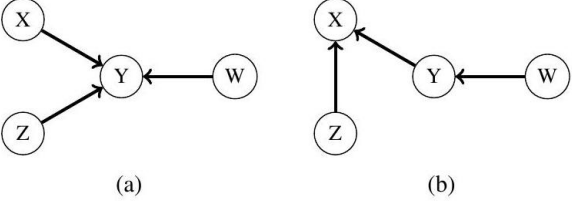
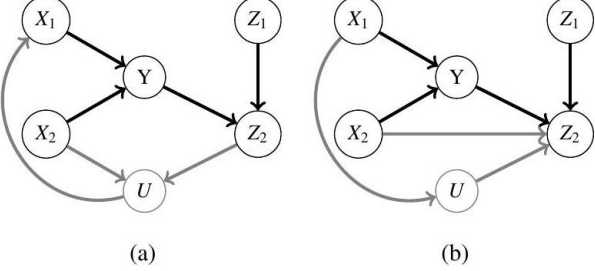
Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00262	011	—	$X_{\pi(j)}$	$X_{\pi(j)}$
2010.14265_F00263	011	—	$\pi^{\{*\}}$	π^*
2010.14265_F00264	011	—	$G_{\pi^{\{*\}}}$	G_{π^*}
2010.14265_F00265	011	—	$\pi(4)=Y$	$\pi(4)=Y$
2010.14265_F00266	011	—	$\pi(1)=X, \pi(2)=Z$	$\pi(1)=X, \pi(2)=Z$
2010.14265_F00267	011	—	$\pi(3)=W$	$\pi(3)=W$
2010.14265_F00268	012	—	$\pi^{\{\prime\}}$	π'
2010.14265_F00269	012	—	$\pi^{\{\prime\}}(3)=Y$	$\pi'(3)=Y$
2010.14265_F00270	012	—	$\pi^{\{\prime\}}(4)=W$	$\pi'(4)=W$
2010.14265_F00271	012	—	$\pi^{\{\prime\}}(1)=Z, \pi^{\{\prime\}}(2)=$	$\pi'(1)=Z, \pi'(2)=$
2010.14265_F00272	012	—	$W, \pi^{\{\prime\}}(3)=Y$	$W, \pi'(3)=Y$
2010.14265_F00273	012	—	$\pi^{\{\prime\}}(4)=X$	$\pi'(4)=X$
2010.14265_F00274	012	—	$\pi^{\{\prime\}}(1)=Y$	$\pi'(1)=Y$
2010.14265_F00275	012	—	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$	$\boldsymbol{W}, \boldsymbol{X}, \boldsymbol{Y}, \boldsymbol{Z} \subseteq \boldsymbol{V}$
2010.14265_F00276	012	—	$\left(\perp\right)$	<i>(not rendered)</i>
2010.14265_F00277	012	—	$\perp_P$	<i>(not rendered)</i>
2010.14265_F00278	012	—	$\left.\perp_G\right)$	<i>(not rendered)</i>
2010.14265_F00279		—	$\boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z} \rightarrow \boldsymbol{Y} \perp \boldsymbol{X} \boldsymbol{Z}$	<i>(not rendered)</i>
2010.14265_F00280		—	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{Z}$	<i>(not rendered)</i>
2010.14265_F00281		—	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \boldsymbol{Z} \rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \boldsymbol{W} \cup \boldsymbol{Z}$	<i>(not rendered)</i>
2010.14265_F00282	012	—	$(\boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \mid \boldsymbol{Z}) \rightarrow$	<i>(not rendered)</i>
2010.14265_F00283	012	—	$\boldsymbol{X} \perp \boldsymbol{Y} \cup \boldsymbol{W} \mid \boldsymbol{Z}$	<i>(not rendered)</i>
2010.14265_F00284	012	—	$\perp_G$	<i>(not rendered)</i>
2010.14265_F00285	012	—	$(\boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{W} \cup \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \mid \boldsymbol{Y} \cup \boldsymbol{Z}) \rightarrow$	<i>(not rendered)</i>
2010.14265_F00286	012	—	$(\boldsymbol{X} \perp \boldsymbol{Y} \mid \boldsymbol{Z}) \wedge (\boldsymbol{X} \perp \boldsymbol{W} \mid \boldsymbol{Z}) \rightarrow \boldsymbol{X} \perp \boldsymbol{Y} \cup$	<i>(not rendered)</i>
2010.14265_F00287	012	—	$\boldsymbol{W} \mid \boldsymbol{Z}$	$\boldsymbol{W} \mid \boldsymbol{Z}$
2010.14265_F00288	012	—	$Y \perp_P X$	<i>(not rendered)</i>
2010.14265_F00289	012	—	$Y \perp_P Z$	<i>(not rendered)</i>
2010.14265_F00290	012	—	$Y \perp_P \{X, Z\}$	<i>(not rendered)</i>
2010.14265_F00291	012	—	$X \not\perp_P \{Y, Z\}, Y \not\perp_P \{X, Z\}$	<i>(not rendered)</i>
2010.14265_F00292	012	—	$X \not\measuredangle_P \{Y, Z\}$	$X \not\measuredangle_P \{Y, Z\}$
2010.14265_F00293	012	—	$Y \perp_P X \mid Z$	<i>(not rendered)</i>
2010.14265_F00294	012	—	$X \perp_P \{Y, Z\}$	<i>(not rendered)</i>
2010.14265_F00295	012	—	$X \not\perp_P Y \mid Z$	<i>(not rendered)</i>
2010.14265_F00296		—	$\square$	$\square$
2010.14265_F00297	005	—	$X \perp_G Y$	<i>(not rendered)</i>
2010.14265_F00298	005	—	$X \perp_G Z$	<i>(not rendered)</i>
2010.14265_F00299	005	—	$X \perp_G \{Y, Z\}$	<i>(not rendered)</i>
2010.14265_F00300	005	—	$X \not\perp_P Y \left Z \cup S, X \not\perp_P Z \right Y \cup S$	<i>(not rendered)</i>
2010.14265_F00301	005	—	$Y \not\perp_P Z \mid X \cup S$	<i>(not rendered)</i>
2010.14265_F00302	013	—	$Z \notin \operatorname{An}(X)$	$Z \notin \operatorname{An}(X)$
2010.14265_F00303	013	—	$Z \notin \operatorname{An}(Y)$	$Z \notin \operatorname{An}(Y)$
2010.14265_F00304	013	—	$X \notin$	$X \notin$
2010.14265_F00305	013	—	$\operatorname{De}(Z)$	$\operatorname{De}(Z)$
2010.14265_F00306	013	—	$Y \notin \operatorname{De}(Z)$	$Y \notin \operatorname{De}(Z)$
2010.14265_F00307	013	—	$Z \perp_G \operatorname{Nd}(Z) \mid \operatorname{Pa}(Z)$	<i>(not rendered)</i>
2010.14265_F00308	013	—	$ \operatorname{Pa}(Z) \cap \{X, Y\} < 2$	$ \operatorname{Pa}(Z) \cap \{X, Y\} < 2$
2010.14265_F00309	013	—	$\{X, Y\} \subseteq \operatorname{Pa}(Z)$	$\{X, Y\} \subseteq \operatorname{Pa}(Z)$
2010.14265_F00310	013	—	$\operatorname{Pa}(X) \subseteq \boldsymbol{V} \backslash \boldsymbol{Z}$	$\operatorname{Pa}(X) \subseteq \boldsymbol{V} \backslash \boldsymbol{Z}$
2010.14265_F00311	013	—	$\boldsymbol{X} \cup \boldsymbol{Z}$	$\boldsymbol{X} \cup \boldsymbol{Z}$
2010.14265_F00312	013	—	$(\boldsymbol{X} \cup \boldsymbol{Z}) \backslash \{X\}$	$(\boldsymbol{X} \cup \boldsymbol{Z}) \backslash \{X\}$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00313	013	—	$(\boldsymbol{X} \cup \boldsymbol{Z}) \backslash \{X\} \subseteq \operatorname{Nd}(X)$	$(\boldsymbol{X} \cup \boldsymbol{Z}) \backslash \{X\} \subseteq \operatorname{Nd}(X)$
2010.14265_F00314	013	—	$X \perp_{\{G\}} (\boldsymbol{X} \cup \boldsymbol{Z}) \backslash \{X\} \mid \operatorname{Pa}(X)$	<i>(not rendered)</i>
2010.14265_F00315	013	—	$Z \cap \operatorname{Pa}(X) = \emptyset$	$Z \cap \operatorname{Pa}(X) = \emptyset$
2010.14265_F00316	013	—	$\boldsymbol{X} \rightarrow \boldsymbol{Y} \leftarrow \boldsymbol{Z}$	$\boldsymbol{X} \rightarrow \boldsymbol{Y} \leftarrow \boldsymbol{Z}$
2010.14265_F00317	013	—	$\boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Z\}$	$\boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, Z\}$
2010.14265_F00318	013	—	$X \perp_{\{G\}} Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\})$	<i>(not rendered)</i>
2010.14265_F00319	013	—	$Y \in$	$Y \in$
2010.14265_F00320	013	—	$\operatorname{Nd}(X)$	$\operatorname{Nd}(X)$
2010.14265_F00321	013	—	$X \perp_{\{G\}} Z \mid \operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup$	<i>(not rendered)</i>
2010.14265_F00322	013	—	$(\boldsymbol{Z} \backslash \{Z\}) \cup \{Y\}$	$(\boldsymbol{Z} \backslash \{Z\}) \cup \{Y\}$
2010.14265_F00323	013	—	$\operatorname{De}(Y)$	$\operatorname{De}(Y)$
2010.14265_F00324	013	—	$Y \in \operatorname{Nd}(X)$	$Y \in \operatorname{Nd}(X)$
2010.14265_F00325	013	—	$X \in \operatorname{Nd}(Y)$	$X \in \operatorname{Nd}(Y)$
2010.14265_F00326	013	—	$Y, X \in \operatorname{Pa}(Y)$	$Y, X \in \operatorname{Pa}(Y)$
2010.14265_F00327	013	—	$X^{\prime}$	$X^{\prime}$
2010.14265_F00328	013	—	$X^{\prime} \in \operatorname{Nd}(X)$	$X^{\prime} \in \operatorname{Nd}(X)$
2010.14265_F00329	013	—	$\left\{X, X^{\prime}, Y\right\}$	$\{X, X^{\prime}, Y\}$
2010.14265_F00330	013	—	$X \notin \operatorname{De}(Y)$	$X \notin \operatorname{De}(Y)$
2010.14265_F00331	013	—	$X \notin \operatorname{An}(X^{\prime})$	$X \notin \operatorname{An}(X^{\prime})$
2010.14265_F00332	013	—	$\langle X, Y, X^{\prime} \rangle$	$\langle X, Y, X^{\prime} \rangle$
2010.14265_F00333	013	—	$X \in \operatorname{An}(Z)$	$X \in \operatorname{An}(Z)$
2010.14265_F00334	013	—	$\operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\})$	$\operatorname{Pa}(X) \cup (\boldsymbol{X} \backslash \{X\}) \cup (\boldsymbol{Z} \backslash \{Z\})$
2010.14265_F00335	008	—	$\boldsymbol{X} \cap \boldsymbol{Z} = \emptyset, Y_{-_ \leq 2} \boldsymbol{X}, Y^s_{_ \leq 2} \boldsymbol{Z}$	$\boldsymbol{X} \cap \boldsymbol{Z} = \emptyset, Y_{-_ \leq 2} \boldsymbol{X}, Y^s_{_ \leq 2} \boldsymbol{Z}$
2010.14265_F00336	014	—	$\operatorname{MB}(T) \subseteq \boldsymbol{S}$	$\operatorname{MB}(T) \subseteq \boldsymbol{S}$
2010.14265_F00337	014	—	$\operatorname{MB}(T)$	$\operatorname{MB}(T)$
2010.14265_F00338	014	—	$X \in \operatorname{PC}(T), T$	$X \in \operatorname{PC}(T), T$
2010.14265_F00339	014	—	$\frac{s^s}{2} \boldsymbol{X}$	$\frac{s^s}{2} \boldsymbol{X}$
2010.14265_F00340	014	—	$T, \perp_{\{P\}} X \mid \boldsymbol{S}$	<i>(not rendered)</i>
2010.14265_F00341	014	—	$X \notin \boldsymbol{S}$	$X \notin \boldsymbol{S}$
2010.14265_F00342	014	—	$T \not\perp_{\{P\}} X \mid \boldsymbol{S} \cup \{Z\}$	<i>(not rendered)</i>
2010.14265_F00343	014	—	$\boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, T, Z\}$	$\boldsymbol{S} \subseteq \boldsymbol{V} \backslash \{X, T, Z\}$
2010.14265_F00344	014	—	$T_{\{\perp_{\{P\}}\}} Z \mid S$	<i>(not rendered)</i>
2010.14265_F00345	014	—	$Z \notin S$	$Z \notin S$
2010.14265_F00346	014	—	$X \in S$	$X \in S$
2010.14265_F00347	014	—	(T)	(T)
2010.14265_F00348	014	—	$S: 1)$	$S: 1)$
2010.14265_F00349	014	—	$T \left(T_{2_2}^s \{C, S\} \right)$	$T \left(T_{2_2}^s \{C, S\} \right)$
2010.14265_F00350	014	—	S	S
2010.14265_F00351	014	—	S_1	S_1
2010.14265_F00352	014	—	S_2	S_2
2010.14265_F00353	014	—	$W \notin \operatorname{MB}(T)$	$W \notin \operatorname{MB}(T)$
2010.14265_F00354	014	—	$X \in \operatorname{MB}(T)$	$X \in \operatorname{MB}(T)$
2010.14265_F00355		—	$\operatorname{PC}(T)$	$\operatorname{PC}(T)$
2010.14265_F00356	014	—	$X \in \operatorname{PC}(T)$	$X \in \operatorname{PC}(T)$
2010.14265_F00357	014	—	$\boldsymbol{X} \subseteq \operatorname{MB}(T)$	$\boldsymbol{X} \subseteq \operatorname{MB}(T)$
2010.14265_F00358	014	—	$\boldsymbol{X} = \{X, Z\}$	$\boldsymbol{X} = \{X, Z\}$
2010.14265_F00359	014	—	$_{\{P\}} X \mid \boldsymbol{S} \backslash \{X\}$	$_{\{P\}} X \mid \boldsymbol{S} \backslash \{X\}$
2010.14265_F00360	014	—	$\operatorname{MB}(T) \backslash \{X\}$	$\operatorname{MB}(T) \backslash \{X\}$
2010.14265_F00361	014	—	$X \perp_{\{P\}} T \mid \boldsymbol{S} \backslash \{X\}$	<i>(not rendered)</i>
2010.14265_F00362	014	—	$W \notin \operatorname{MB}(T)$	$W \notin \operatorname{MB}(T)$
2010.14265_F00363	014	—	$W \in S$	$W \in S$
2010.14265_F00364	014	—	$\boldsymbol{S} \backslash \{W\}$	$\boldsymbol{S} \backslash \{W\}$
2010.14265_F00365	014	—	$\operatorname{MB}(T) \cup \boldsymbol{Q}$	$\operatorname{MB}(T) \cup \boldsymbol{Q}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Identifier	Page	Conf.	LaTeX source	Rendered
2010.14265_F00366	014	—	$\boldsymbol{Q}$	Q
2010.14265_F00367	014	—	$T \perp_{\{G\}} \{W\} \cup \boldsymbol{Q} \mid \operatorname{MB}(T)$	<i>(not rendered)</i>
2010.14265_F00368	014	—	$T \perp_{\{G\}} W \mid \operatorname{MB}(T) \cup \boldsymbol{Q}$	<i>(not rendered)</i>
2010.14265_F00369	014	—	$T \perp_{\{P\}} W \mid \boldsymbol{S} \backslash W$	<i>(not rendered)</i>
Conf. MathPix confidence: ■ ≥ 0.9 ■ 0.5–0.9 ■ < 0.5 — no value Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured				

Unrecovered image regions — TikZ / table / failed-math candidates

Regions MathPix left as images (no LaTeX). Reconstruct with pdfdrill vision (LLM classify + transcribe to TikZ/tabular/math); verify an LLM result against the real ink with inkdrill.

Identifier	Page	Image	Source
2010.14265_DIA_0001	001		tex.zip (local)
2010.14265_DIA_0002	003		tex.zip (local)
2010.14265_DIA_0003	005		tex.zip (local)
2010.14265_DIA_0004	006		tex.zip (local)
2010.14265_DIA_0005	007		tex.zip (local)
2010.14265_DIA_0006	009	(image not on disk — pass -texzip or download the CDN crop)	CDN only